

Electromagnetic Theory & Quantum Physics

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Prerequisite Knowledge

- Comfort with exponential and logarithmic functions
- Definite integrals (see the Calculus handout — one integral here is simply quoted, since its full evaluation needs contour integration, but everything else is derived directly)
- Basic idea of temperature as a measure of average thermal energy (helpful, not required)

1 The Electromagnetic Spectrum

What Light Actually Is

Here's a fact that's easy to state and genuinely strange to sit with: light is not a stream of tiny bullets, and it's not a disturbance rippling through some invisible material medium (the 19th-century "aether," which turned out not to exist). Light is a self-sustaining disturbance in the electric and magnetic fields themselves — a changing electric field creates a changing magnetic field, which in turn creates a changing electric field, and this hand-off repeats forever, propagating outward through empty space at a fixed speed c , needing nothing at all to "wave" in.

You don't need the full machinery of Maxwell's equations to appreciate the punchline: this self-sustaining wave has a wavelength λ and a frequency f , related to its speed by the same relationship that governs any wave (water waves, sound waves, the wave on a guitar string):

$$c = f\lambda.$$

Since c is a fixed constant for light in vacuum, wavelength and frequency are just two different labels for the same underlying property — know one, and you immediately know the other. A "short wavelength" wave is automatically a "high frequency" wave, and vice versa.

Classifying the Spectrum

The **electromagnetic spectrum** is simply the full range of possible wavelengths (or equivalently, frequencies) that this kind of wave can have — and it turns out visible light, the narrow sliver our eyes happen to respond to, is a tiny fraction of what's actually out there. Going from long wavelength (low frequency, low energy) to short wavelength (high frequency, high energy):

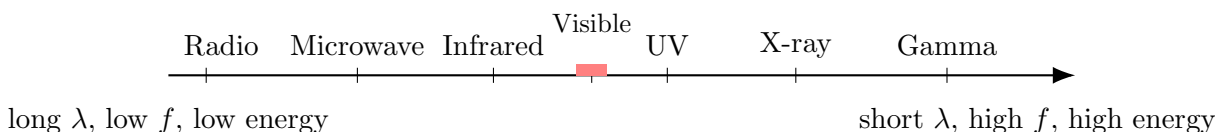


Figure 1: The electromagnetic spectrum. The tiny red sliver marks visible light — roughly 400–700 nanometres, an almost embarrassingly small window onto everything that's actually happening electromagnetically in the universe.

A rough sense of scale is worth having memorized: radio waves can be kilometres long; visible light sits around 400–700 nanometres ($4\text{--}7 \times 10^{-7}$ m); X-rays are comparable to the size of atoms; and gamma rays are smaller still, comparable to atomic nuclei. This is precisely why different wavelengths interact so differently with matter — a wave's ability to resolve or be blocked by an object depends on how the wavelength compares to the object's own size, a theme that will resurface directly when we get to telescope resolution elsewhere in your prep.

Why Astronomers Care Which Windows Are Open

Earth's atmosphere is not transparent to all of this spectrum — far from it. Only a couple of "windows" let radiation through cleanly to the ground: the **optical window** (roughly visible light, plus a little near-infrared and near-ultraviolet) and the **radio window** (a broad swath of radio wavelengths). Almost everything else — most infrared, ultraviolet, X-ray, and gamma-ray radiation — is absorbed high in the atmosphere, mostly by water vapor, ozone, and the ionosphere.

This is precisely why X-ray and gamma-ray telescopes have to be launched into space, while radio and optical telescopes can comfortably sit on the ground: it's not a matter of technology, but of which photons ever survive the trip down through our own air.

2 Radiation Laws

Setting the Stage: Why Hot Things Glow

Every object with a temperature above absolute zero is made of charged particles (electrons, protons) that are jiggling around due to thermal energy. Accelerating charges radiate electromagnetic waves — that's a basic fact of electromagnetism, and it means *every* warm object is constantly emitting light, across a whole range of wavelengths, purely because it's warm. A red-hot stove, your own body (emitting mostly in the infrared, which is why night-vision goggles work), and the Sun are all doing fundamentally the same thing — just at wildly different temperatures, which is why they glow in such different colors (or don't visibly glow at all).

The two laws in this section describe, respectively, *where* the emission peaks (which color dominates) and *how much* total power comes out. Both depend on nothing but temperature — not on what the object is made of, not on its size, just T . That universality is itself a clue that something deep is going on, and we'll get to the "why" properly in the next section.

Wien's Displacement Law

Empirically (and, as we'll derive properly once we have Planck's law in hand), the wavelength at which a hot object's emission is strongest is inversely proportional to its temperature:

$$\boxed{\lambda_{\max} T = b}, \quad b \approx 2.898 \times 10^{-3} \text{ m}\cdot\text{K}$$

(Wien's constant). The intuition: hotter objects have more thermal energy available per particle, which lets them shake out photons of correspondingly higher energy (shorter wavelength) more readily. A cool object like a human body ($\approx 300 \text{ K}$) peaks deep in the infrared, invisible to our eyes; a very hot star ($\approx 30,000 \text{ K}$) peaks in the ultraviolet, appearing blue-white to us because we're only catching the visible tail of a curve that's actually peaking somewhere we can't see at all.

The Stefan-Boltzmann Law

The *total* power radiated per unit surface area, summed over all wavelengths, scales as the fourth power of temperature:

$$\boxed{F = \sigma T^4}, \quad \sigma \approx 5.67 \times 10^{-8} \text{ W m}^{-2}\text{K}^{-4}.$$

That T^4 scaling is dramatic — doubling an object's absolute temperature increases its total radiated power per unit area by a factor of 16. For a spherical star of radius R , multiplying by the total surface area gives the star's luminosity:

$$\boxed{L = 4\pi R^2 \sigma T^4}.$$

This single equation is the workhorse connecting three of the most fundamental observable properties of a star — its luminosity, radius, and surface temperature — and it's worth having completely automatic.

Worked Example — The Sun’s Surface Temperature

The mean solar flux measured at Earth (the "solar constant") is $F_{\oplus} \approx 1.36 \times 10^3 \text{ W/m}^2$, at a distance $d = 1.5 \times 10^{11} \text{ m}$. Since flux falls off as $1/d^2$ (the same total power spreading over an ever-larger sphere as it travels outward), the Sun’s total luminosity is

$$L_{\odot} = F_{\oplus} \cdot 4\pi d^2 \approx 3.85 \times 10^{26} \text{ W}.$$

Using $L_{\odot} = 4\pi R_{\odot}^2 \sigma T^4$ with $R_{\odot} \approx 6.96 \times 10^8 \text{ m}$, solving for T :

$$T = \left(\frac{L_{\odot}}{4\pi R_{\odot}^2 \sigma} \right)^{1/4} \approx 5,780 \text{ K},$$

matching the Sun’s known surface temperature closely — a satisfying check that a law derived from pure thermodynamics correctly predicts something measured entirely independently.

3 Blackbody Radiation

What Makes a “Blackbody” Special

A **blackbody** is an idealized object that absorbs *all* radiation falling on it (no reflection at all, hence "black") and, in thermal equilibrium, re-emits radiation with a spectrum that depends *only* on its temperature — not on what it’s made of. The classic physical realization is a cavity with a small hole: any light that wanders in through the hole bounces around inside so many times, being partially absorbed at each bounce, that essentially none of it ever finds its way back out unabsorbed. The hole itself, therefore, is a nearly perfect absorber — and by the same token, radiation streaming *out* of that hole is as close to pure blackbody emission as you can physically arrange.

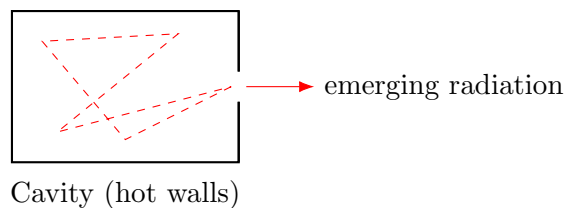


Figure 2: The cavity-with-a-hole realization of a blackbody. Light entering the hole bounces around many times, getting absorbed and re-emitted by the hot walls at each bounce, before a fraction eventually finds its way back out — the emerging radiation is a nearly perfect sample of the cavity’s thermal equilibrium spectrum.

Stars are, to an excellent approximation, blackbodies — not because they’re literal cavities, but because light generated deep inside gets absorbed and re-emitted so many times on its way out through the dense stellar material that, by the time it escapes, it has thoroughly "forgotten" its original details and carries only the imprint of the local temperature.

The Puzzle That 19th-Century Physics Couldn’t Solve

Before quantum mechanics, physicists tried to predict the blackbody spectrum using classical statistical mechanics, via a principle called **equipartition**: in thermal equilibrium, every independent

way a system can store energy (every "mode") gets, on average, the same share of thermal energy, kT (up to a fixed numerical factor). Applied to the electromagnetic field inside a cavity, this predicts that *every* possible wave mode — no matter how short its wavelength — carries the same average energy kT .

The trouble is that a cavity supports *far more* possible short-wavelength modes than long-wavelength ones (there's simply more geometric room to fit oscillation patterns as the wavelength shrinks), and if every single one of those countless short-wavelength modes carries energy kT , the total predicted energy diverges to infinity. This absurd prediction — that any hot object should radiate infinite energy, mostly at the shortest (ultraviolet and beyond) wavelengths — became notorious as the **ultraviolet catastrophe**. Real hot objects obviously don't do this; something about the classical reasoning had to be wrong.

Planck's Fix: Quantized Energy

Max Planck's resolution, initially treated as more of a mathematical trick than a physical claim, was to propose that an oscillator of frequency f can't hold just *any* amount of energy — it can only hold energy in discrete lumps,

$$E_n = nhf, \quad n = 0, 1, 2, 3, \dots,$$

where $h \approx 6.626 \times 10^{-34}$ J·s is Planck's constant. This single assumption, as we're about to see, completely cures the ultraviolet catastrophe.

Deriving the Average Energy of a Quantized Oscillator

In thermal equilibrium at temperature T , the probability of an oscillator occupying energy level E_n follows the Boltzmann distribution, weighted by $e^{-E_n/kT}$. The average energy is then a weighted sum over all the levels:

$$\langle E \rangle = \frac{\sum_{n=0}^{\infty} nhf e^{-nhf/kT}}{\sum_{n=0}^{\infty} e^{-nhf/kT}}.$$

Let $x \equiv e^{-hf/kT}$ (a number between 0 and 1) to simplify the bookkeeping. The denominator is a standard geometric series:

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}.$$

The numerator needs one more trick: differentiate the geometric series with respect to x (term-by-term, which is legitimate here) and multiply by x :

$$\sum_{n=0}^{\infty} nx^n = x \frac{d}{dx} \left(\frac{1}{1-x} \right) = \frac{x}{(1-x)^2}.$$

Putting the pieces together:

$$\langle E \rangle = hf \cdot \frac{x/(1-x)^2}{1/(1-x)} = \frac{hfx}{1-x} = \frac{hf}{x^{-1} - 1}.$$

Substituting back $x^{-1} = e^{hf/kT}$:

$$\langle E \rangle = \frac{hf}{e^{hf/kT} - 1}.$$

Why This Cures the Catastrophe

Look at the two limits of this formula. When $hf \ll kT$ (low frequency, or equivalently high temperature), expand $e^{hf/kT} \approx 1 + hf/kT$, giving $\langle E \rangle \approx kT$ — recovering the old classical equipartition result exactly, as it must, since quantization shouldn't matter when the energy quantum hf is tiny compared to the available thermal energy kT .

But when $hf \gg kT$ (high frequency), the denominator $e^{hf/kT}$ becomes astronomically large, and $\langle E \rangle \rightarrow 0$ exponentially fast. Physically: a high-frequency mode requires at least one full quantum hf to be excited at all, and if that quantum is much larger than the typical thermal energy kT sloshing around, the odds of an oscillator ever accumulating that much energy become vanishingly small. High-frequency modes are, in effect, "frozen out" — exactly the suppression needed to avoid the ultraviolet catastrophe, and it falls directly out of treating energy as discrete rather than continuous.

Assembling Planck's Law

To get the full spectrum, we need one more ingredient: how many electromagnetic modes actually exist per unit frequency interval, per unit volume, inside a cavity. Counting the standing-wave patterns that fit inside a 3D box (a geometric exercise we won't carry out in full here) gives a mode density that grows as f^2 :

$$g(f) df = \frac{8\pi f^2}{c^3} df.$$

Multiplying this mode density by the average energy per mode we just derived gives the full spectral energy density — **Planck's Law**:

$$u(f, T) df = \frac{8\pi f^2}{c^3} \cdot \frac{hf}{e^{hf/kT} - 1} df.$$

This single formula — built from nothing but mode-counting and quantized thermal averaging — correctly predicts the entire observed blackbody spectrum, at every frequency and every temperature, with no free parameters once h is fixed. It was, historically, the single equation that forced physicists to take energy quantization seriously as a fact about nature, not just a convenient mathematical fudge.

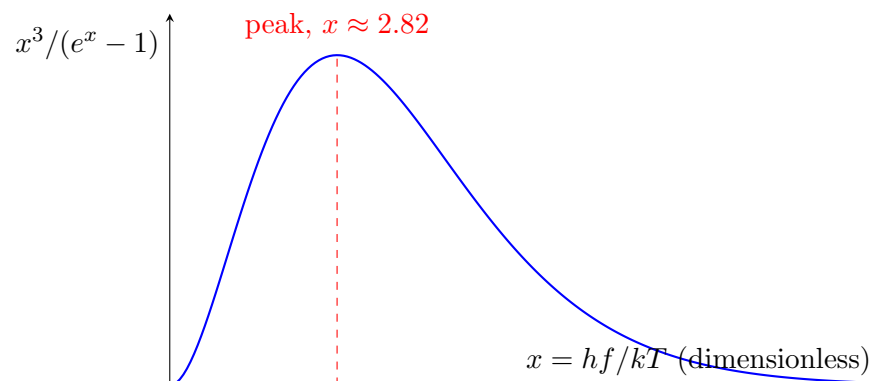


Figure 3: The universal, dimensionless shape of the Planck spectrum (relative intensity $x^3/(e^x - 1)$), which is exactly Planck's law rewritten with $x = hf/kT$). Every blackbody, at every temperature, produces this same curve — only the horizontal scaling by kT/h changes, which is precisely the content of Wien's law.

Deriving Wien's Law From Planck's Law

Wien's law isn't a separate physical assumption — it's a direct consequence of Planck's law, found by locating the peak of the curve above. Working in terms of $x = hf/kT$, the curve's shape is $x^3/(e^x - 1)$ (up to overall constants that don't affect where the maximum sits). Setting the derivative to zero:

$$\frac{d}{dx} \left[\frac{x^3}{e^x - 1} \right] = 0 \implies 3x^2(e^x - 1) - x^3e^x = 0 \implies (3 - x)e^x = 3.$$

This is a transcendental equation — it can't be solved in closed form — but it can be solved numerically, giving $x \approx 2.82$. Converting back, the frequency of peak emission satisfies $hf_{\max}/kT \approx 2.82$, i.e. $f_{\max} \propto T$. (If you instead maximize with respect to *wavelength* rather than frequency, being careful about the change of variables, you get a slightly different numerical constant, $x \approx 4.965$, and $\lambda_{\max}T = b$ — the two versions of Wien's law, for frequency and for wavelength, are both valid but use different constants precisely because f and λ aren't linearly related, so "the peak in frequency" and "the peak in wavelength" are genuinely different points on the same curve.)

Deriving the Stefan-Boltzmann Law From Planck's Law

Finally, let's get the T^4 scaling directly, by integrating Planck's law over all frequencies to get the total energy density:

$$u(T) = \int_0^\infty \frac{8\pi f^2}{c^3} \cdot \frac{hf}{e^{hf/kT} - 1} df = \frac{8\pi h}{c^3} \int_0^\infty \frac{f^3}{e^{hf/kT} - 1} df.$$

Substitute $x = hf/kT$ (so $f = xkT/h$, $df = (kT/h) dx$):

$$u(T) = \frac{8\pi h}{c^3} \left(\frac{kT}{h} \right)^4 \int_0^\infty \frac{x^3}{e^x - 1} dx.$$

Every factor of T has now been cleanly pulled outside the integral, leaving a purely numerical integral that doesn't depend on temperature at all. This particular integral is a well-known result (evaluated using the same techniques that give the Basel problem and its generalizations; we'll just quote it here rather than derive it, since it needs an expansion in terms of the Riemann zeta function):

$$\int_0^\infty \frac{x^3}{e^x - 1} dx = \frac{\pi^4}{15}.$$

Substituting back:

$$u(T) = \frac{8\pi h}{c^3} \cdot \frac{k^4 T^4}{h^4} \cdot \frac{\pi^4}{15} = \frac{8\pi^5 k^4}{15h^3 c^3} T^4.$$

Every constant here (h , k , c) is a fixed number, so we've shown, from first principles, that the total radiated energy density scales as T^4 — exactly the Stefan-Boltzmann law, with the constant σ expressible entirely in terms of more fundamental constants. This is one of the most satisfying moments in this whole handout: a law that looked like an empirical curve-fit ("power goes as the fourth power of temperature, for some reason") turns out to be an exact, derivable consequence of energy quantization plus basic mode-counting, with every single constant traceable back to h , k , and c .

4 Stellar Applications

Everything derived so far is, on its own, just thermal physics — it doesn't know or care whether the hot object in question is a light bulb, a furnace, or a star. But this is precisely what makes it so useful in astronomy: stars are, to excellent approximation, blackbodies, so every law from the last two sections applies to them directly, and turns light itself into a measuring instrument.

Turning Starlight Into a Temperature

Wien's law hands you a star's surface temperature almost for free: measure the wavelength at which a star's spectrum peaks (or, more crudely, its overall color — reddish stars are cooler, bluish stars are hotter), and

$$T = \frac{b}{\lambda_{\max}}.$$

This is why color is such a direct, physically meaningful diagnostic in astronomy, rather than just an aesthetic detail: a star's color *is* a temperature reading, taken from millions of kilometres away, using nothing but the light that happens to reach your telescope.

Turning Starlight Into a Radius

Once you know a star's temperature, and separately know its luminosity, the Stefan-Boltzmann law $L = 4\pi R^2 \sigma T^4$ can be inverted to get the star's radius:

$$R = \sqrt{\frac{L}{4\pi\sigma T^4}}.$$

This is a genuinely remarkable thing to be able to do. We cannot physically reach a star, and even the nearest stars appear as unresolved points of light in all but the most specialized instruments — yet by combining a temperature (from color/Wien's law) with a luminosity (from measured flux and a separately known distance, via the inverse-square law), we can pin down the physical size of an object we will never directly see the disk of.

Worked Example — Full Characterization of a Star

A star has a measured flux at Earth of $F_{\oplus} = 6.23 \times 10^{-4} \text{ W/m}^2$, and is known (from parallax) to be at a distance of $d = 13.9 \text{ pc} \approx 4.29 \times 10^{17} \text{ m}$. Its spectrum peaks at $\lambda_{\max} \approx 1.0 \mu\text{m}$.

Temperature, from Wien's law: $T = b/\lambda_{\max} \approx (2.898 \times 10^{-3})/(1.0 \times 10^{-6}) \approx 2,900 \text{ K}$ (a cool, red star).

Luminosity, from flux and distance: $L = F_{\oplus} \cdot 4\pi d^2 \approx 1.45 \times 10^{26} \text{ W} \approx 0.38 L_{\odot}$.

Radius, from Stefan-Boltzmann: $R = \sqrt{L/(4\pi\sigma T^4)}$, which works out to several times $R_{\odot}$ — telling us this is a cool but comparatively large star, i.e., a red giant rather than a small, dim, cool main-sequence star. Notice that no single measurement told us this on its own; it's only by combining flux, distance, and spectral peak together that the star's true nature emerges.

The HR Diagram: Stefan-Boltzmann's Law, Turned Into a Map

Astronomers plot stars on a diagram of luminosity (vertical axis, often logarithmic) against surface temperature (horizontal axis, logarithmic, and conventionally plotted *backwards*, decreasing to the

right) — the Hertzsprung-Russell (HR) diagram. Stefan-Boltzmann’s law tells you exactly what a line of constant radius looks like on this plot: taking logarithms of $L = 4\pi R^2 \sigma T^4$,

$$\log L = 2 \log R + 4 \log T + \text{const.}$$

At fixed R , this is a straight line in $(\log T, \log L)$ space with slope 4. Stars of different sizes simply live on different parallel diagonal lines, all with that same slope — which is why giant stars (large R) can be simultaneously cool *and* extremely luminous (they sit high up and to the right), while white dwarfs (tiny R) can be hot *and* very dim (low and to the left) — their temperature alone doesn’t determine their luminosity; the diagonal line they sit on does.

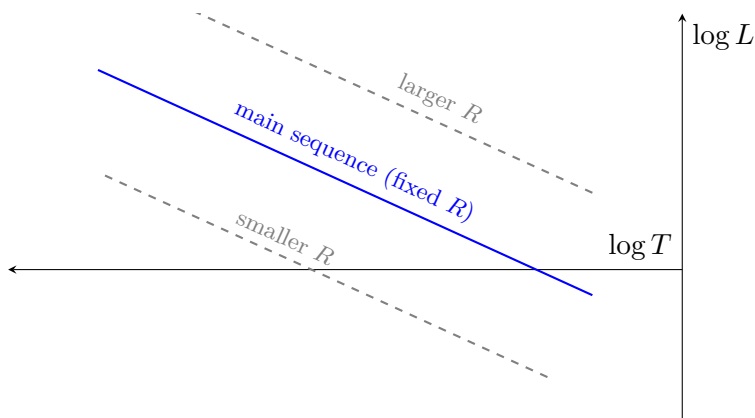


Figure 4: Lines of constant radius on the HR diagram are straight, with slope 4 in $\log L$ vs. $\log T$ — a direct visual signature of the Stefan-Boltzmann law. Note temperature conventionally increases to the left.

Real Stars Aren’t Perfect Blackbodies — and That’s Useful Too

A star’s overall spectral *shape* (the continuum) matches a blackbody curve remarkably well, which is why Wien’s and Stefan-Boltzmann’s laws work at all. But superimposed on that smooth continuum are sharp absorption lines, where atoms and ions in the star’s cooler outer atmosphere absorb light at very specific wavelengths on its way out. The pattern and strength of these lines is what lets astronomers assign a spectral class (O, B, A, F, G, K, M, running from hottest to coolest), which is really just a temperature classification in disguise — hot O and B stars show strong ionized helium lines (needing extreme temperatures to produce), while cool M stars show molecular bands that would be destroyed at higher temperatures. The continuum tells you the temperature via Wien’s law; the line pattern independently confirms (and refines) that same temperature via which atomic transitions are actually possible at that thermal energy.

Worked Example — A Multi-Select Style Question

Two blackbodies, A and B, emit total power in the ratio $L_A : L_B = 2 : 5$. What combinations of temperature ratio and radius ratio are consistent with this?

From $L = 4\pi R^2 \sigma T^4$, the ratio gives one equation in two unknowns:

$$\frac{L_B}{L_A} = \left(\frac{R_B}{R_A}\right)^2 \left(\frac{T_B}{T_A}\right)^4 = \frac{5}{2}.$$

This is deliberately underdetermined — a single equation can’t pin down both ratios independently. So the question becomes: which proposed $(T_B/T_A, R_B/R_A)$ pairs actually satisfy

this equation? For instance, if B is 20% hotter ($T_B/T_A = 1.2$) and 10% larger in diameter ($R_B/R_A = 1.1$), check: $(1.1)^2(1.2)^4 \approx 1.21 \times 2.07 \approx 2.51$, not 2.5 — close, but worth verifying precisely rather than eyeballing, since these problems are built exactly to reward careful substitution over pattern-matching a "plausible-looking" answer.

5 Summary Sheet

Topic	Key Result
Wave relation	$c = f\lambda$
Wien's displacement law	$\lambda_{\max}T = b \approx 2.898 \times 10^{-3} \text{ m}\cdot\text{K}$
Stefan-Boltzmann law	$F = \sigma T^4$; $L = 4\pi R^2 \sigma T^4$
Quantized energy levels	$E_n = nhf$
Average energy of a mode	$\langle E \rangle = \frac{hf}{e^{hf/kT} - 1}$
Mode density in a cavity	$g(f) df = \frac{8\pi f^2}{c^3} df$
Planck's Law	$u(f, T) df = \frac{8\pi f^2}{c^3} \cdot \frac{hf}{e^{hf/kT} - 1} df$
Peak of Planck curve	$x = hf/kT \approx 2.82$ (frequency form)
Stefan-Boltzmann, derived	$u(T) \propto T^4$, using $\int_0^\infty \frac{x^3}{e^x - 1} dx = \pi^4/15$
Stellar temperature	$T = b/\lambda_{\max}$ (Wien's law, applied to a star's spectral peak)
Stellar radius	$R = \sqrt{L/(4\pi\sigma T^4)}$
HR diagram, constant R lines	slope 4 in $\log L$ vs. $\log T$